1D CNN — Architecture Variant Comparison

Generated 2026-05-17 17:47:03

Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic
Total trials	360 (train pool 324, hold-out 36)
Classes	6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap
Sensor channels	144 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	7 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

Sensor grid (✓ enabled, ✗ disabled, — not present)

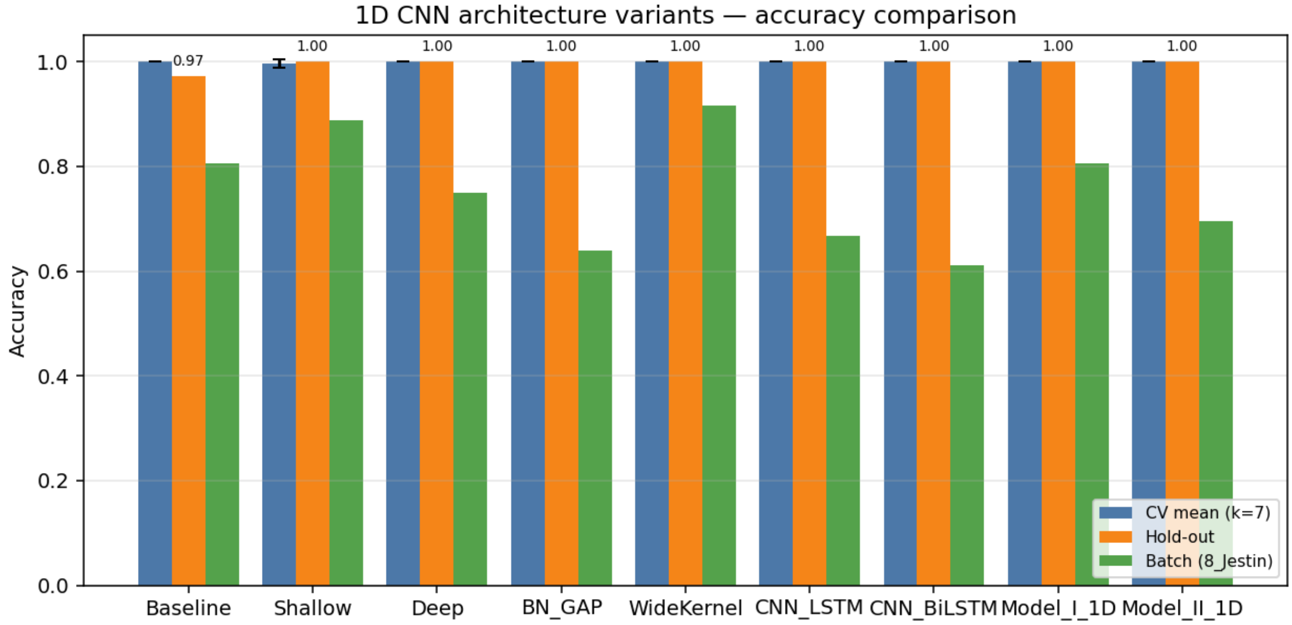
Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✓	✓	✓	✓
left index	✓	✓	✓	✓
left middle	✓	✓	✓	✓
left ring	✓	✓	✓	✓
left pinky	✓	✓	✓	✓
left wrist	✓	—	—	—
right palm	✓	—	—	—
right thumb	✓	✓	✓	✓
right index	✓	✓	✓	✓
right middle	✓	✓	✓	✓
right ring	✓	✓	✓	✓
right pinky	✓	✓	✓	✓
right wrist	✓	—	—	—

Enabled: 44 · Disabled: 0 · Modalities: YPR, Accel → 144 channels total.

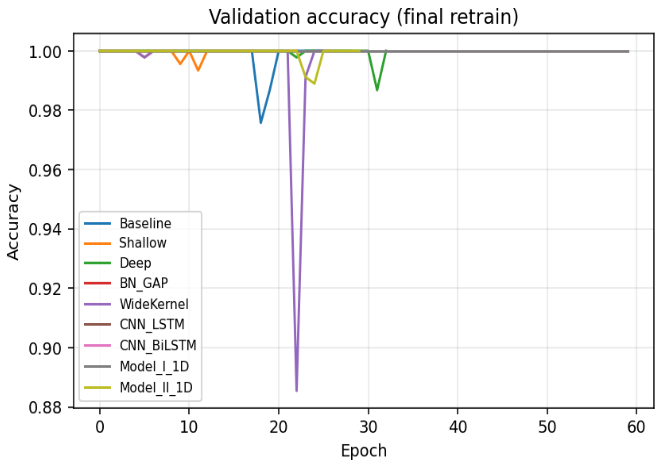
Variant comparison (summary)

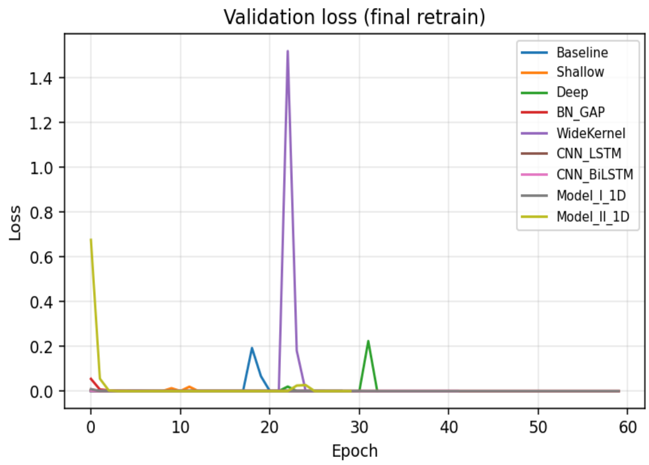
Variant	Params	CV mean (k=7)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	109,094	1.0000	0.0000	0.9722	0.2653	0.8056	29/36
Shallow	67,334	0.9969	0.0076	1.0000	0.0000	0.8889	32/36
Deep	199,782	1.0000	0.0000	1.0000	0.0000	0.7500	27/36
BN_GAP	53,094	1.0000	0.0000	1.0000	0.0000	0.6389	23/36
WideKernel	123,430	1.0000	0.0000	1.0000	0.0000	0.9167	33/36
CNN_LSTM	64,294	1.0000	0.0000	1.0000	0.0000	0.6667	24/36
CNN_BiLSTM	101,414	1.0000	0.0000	1.0000	0.0000	0.6111	22/36
Model_I_1D	118,438	1.0000	0.0000	1.0000	0.0000	0.8056	29/36
Model_II_1D	93,766	1.0000	0.0000	1.0000	0.0000	0.6944	25/36

Best on hold-out: **Shallow** · Best on CV: **Baseline**



Training curves (final retrain on full training pool)





Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	109,094
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	0.9722 · 0.2653
Batch-test acc	0.8056 (29/36)

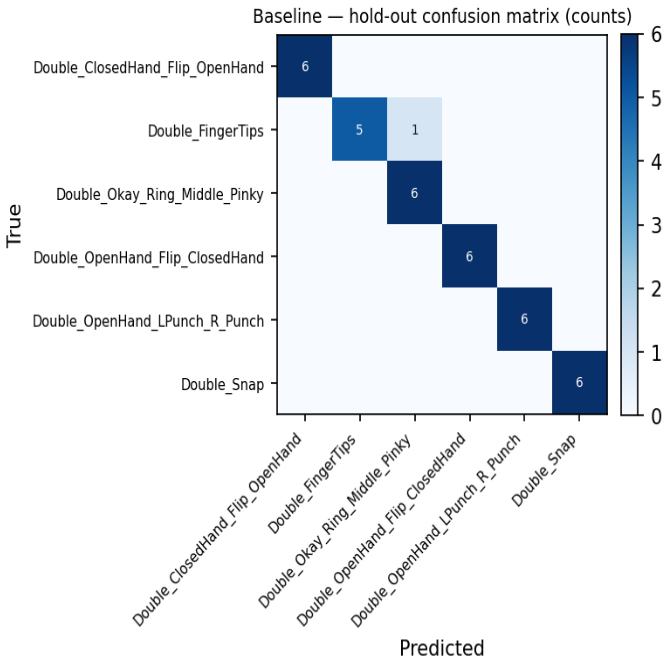
Cross-validation folds

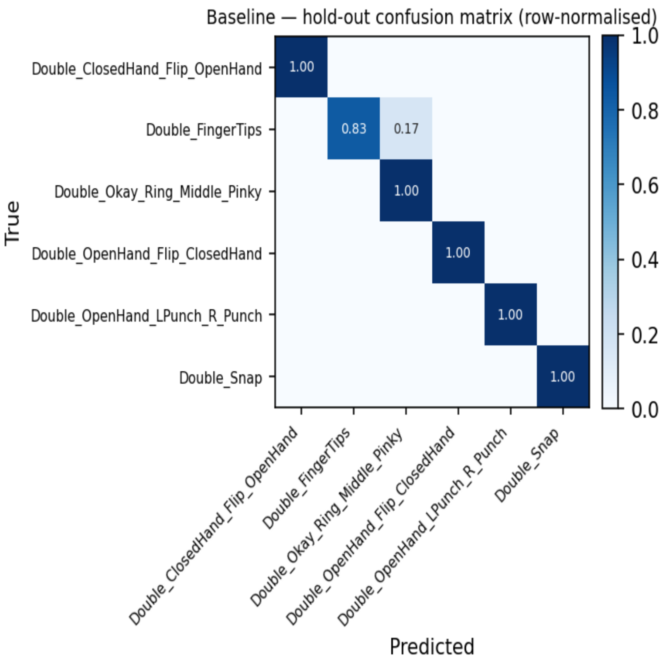
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	0.833	0.909	6
Double_Okay_Ring_Middle_Pinky	0.857	1.000	0.923	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*0.976*	*0.972*	*0.972*	*36*
weighted avg	*0.976*	*0.972*	*0.972*	*36*

Hold-out confusion matrix

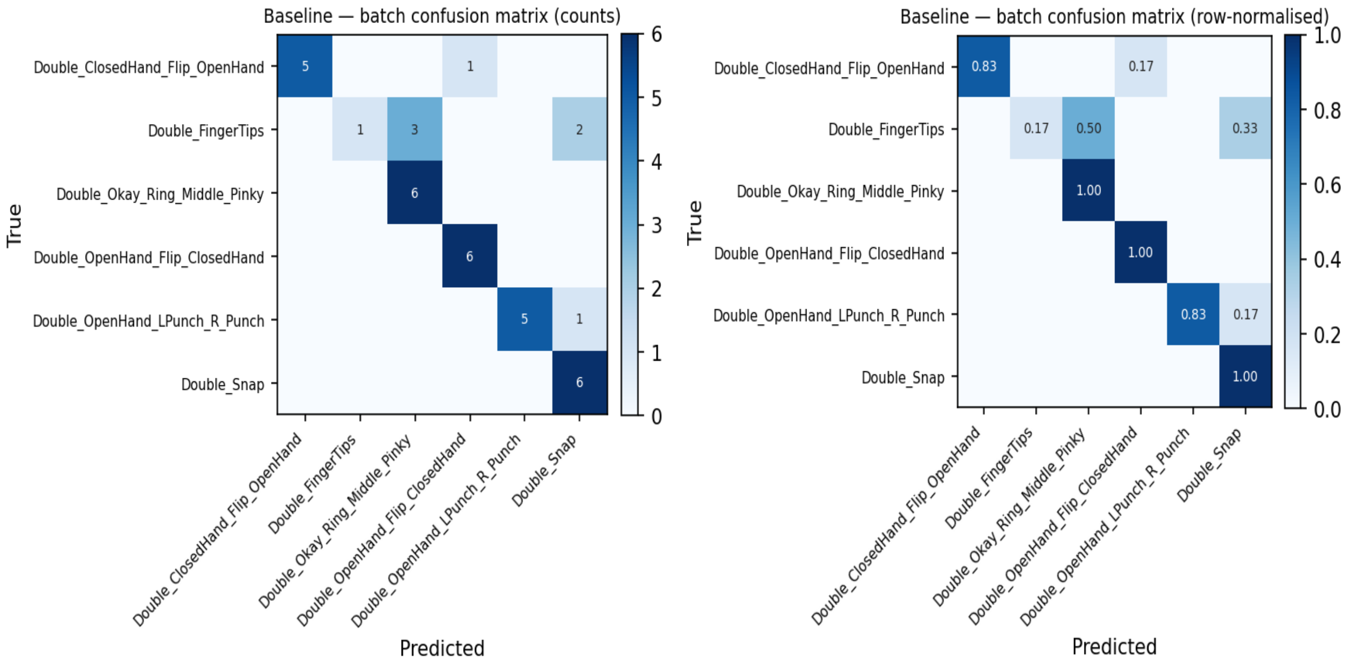




Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	1	6	0.167
Double_Okay_Ring_Middle_Pinky	6	6	1.000
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	5	6	0.833
Double_Snap	6	6	1.000
Overall	**29**	**36**	**0.806**

Batch confusion matrix (8_Jestin)



Baseline — batch confusion matrix (counts)
Baseline — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	67,334
CV mean ± std	0.9969 ± 0.0076
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.8889 (32/36)

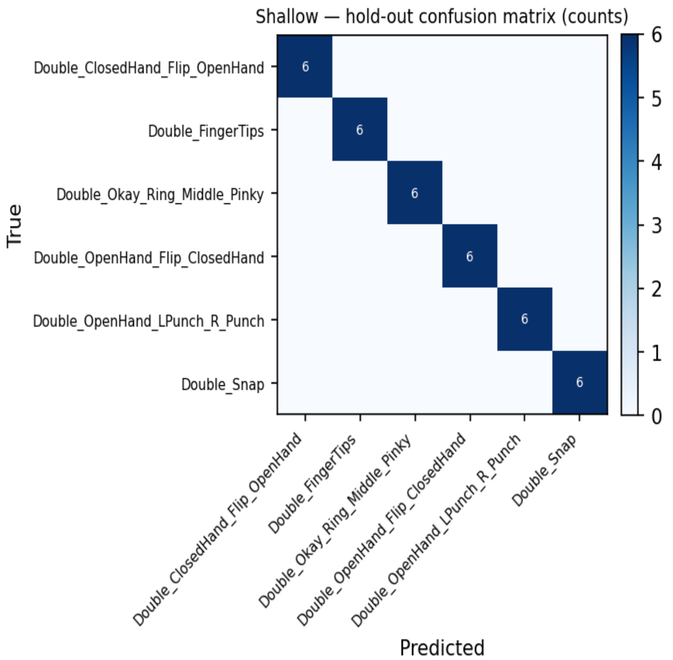
Cross-validation folds

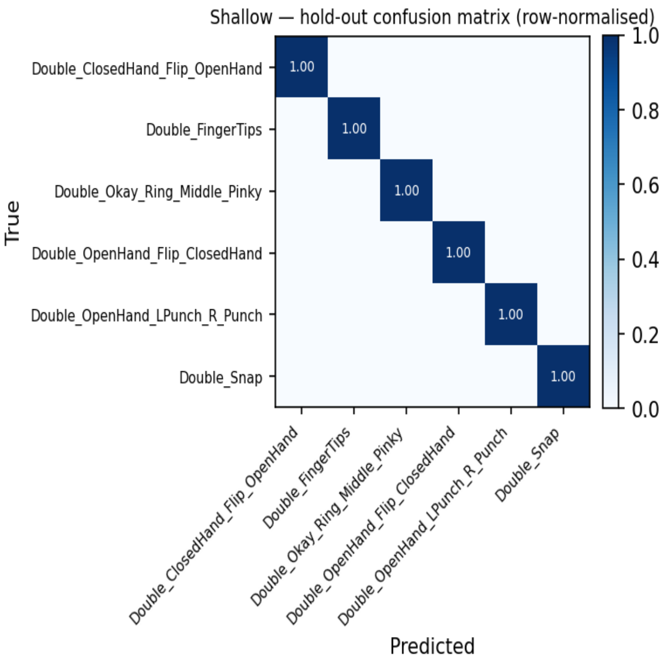
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	0.9783	0.9783	0.9783

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

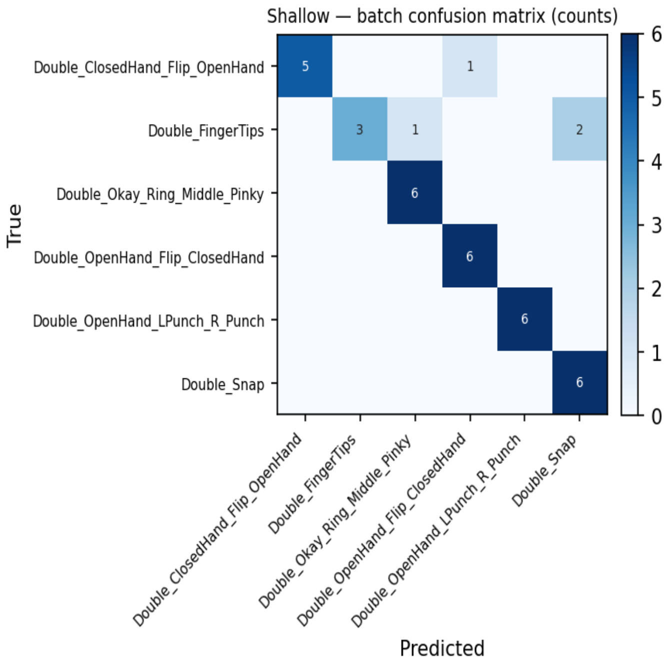




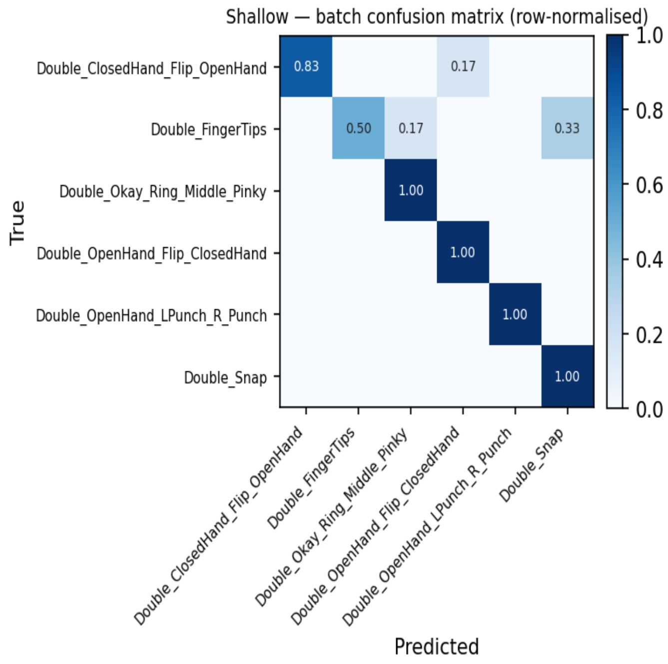
Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	3	6	0.500
Double_Okay_Ring_Middle_Pinky	6	6	1.000
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	6	6	1.000
Double_Snap	6	6	1.000
Overall	**32**	**36**	**0.889**

Batch confusion matrix (8_Jestin)



Shallow — batch confusion matrix (counts)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap



Shallow — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	199,782
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.7500 (27/36)

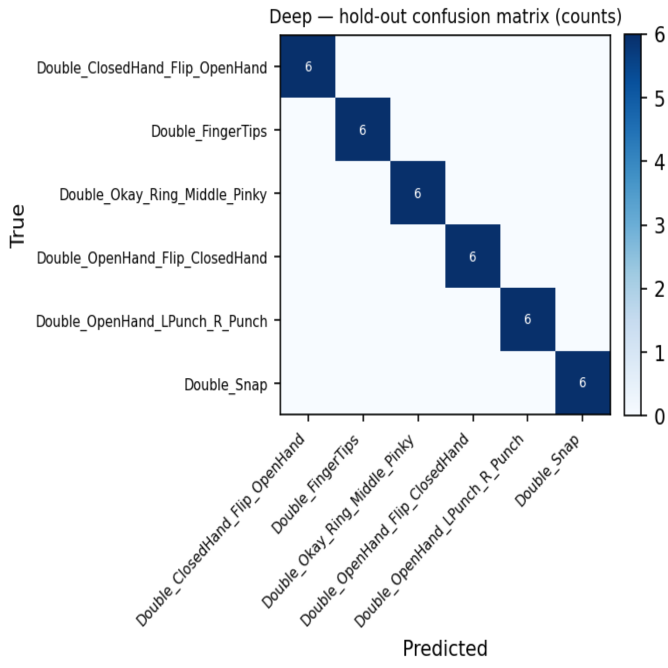
Cross-validation folds

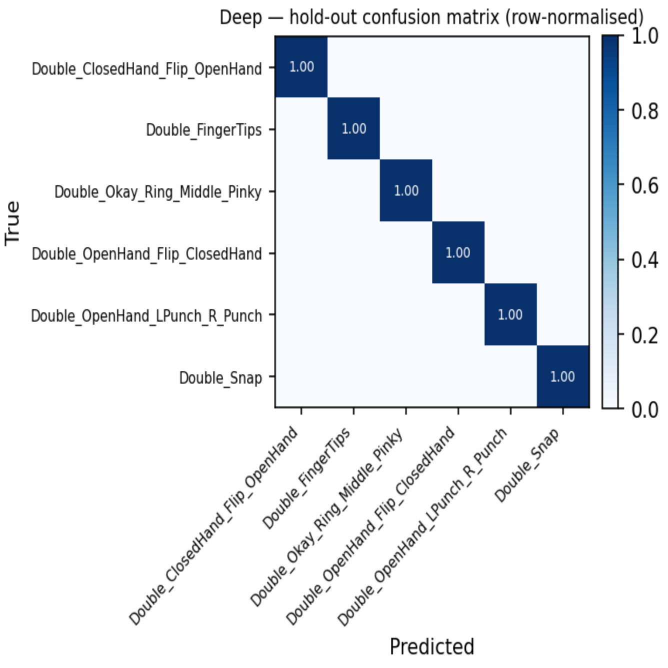
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

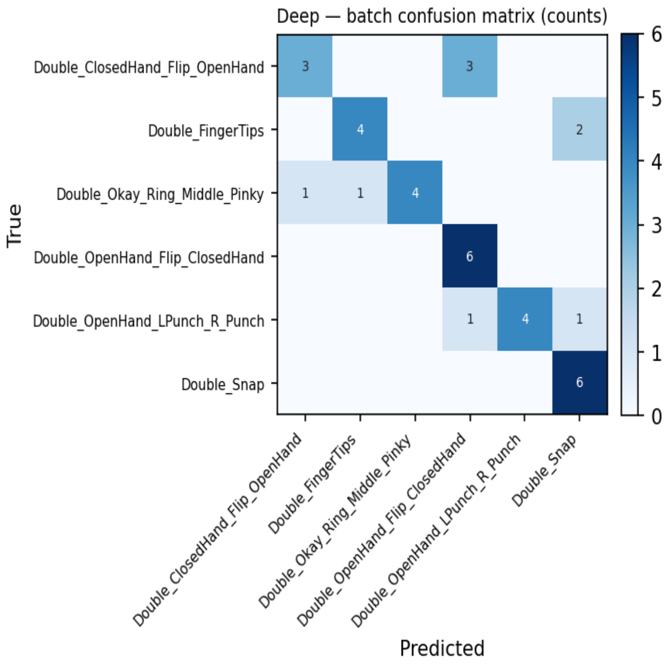




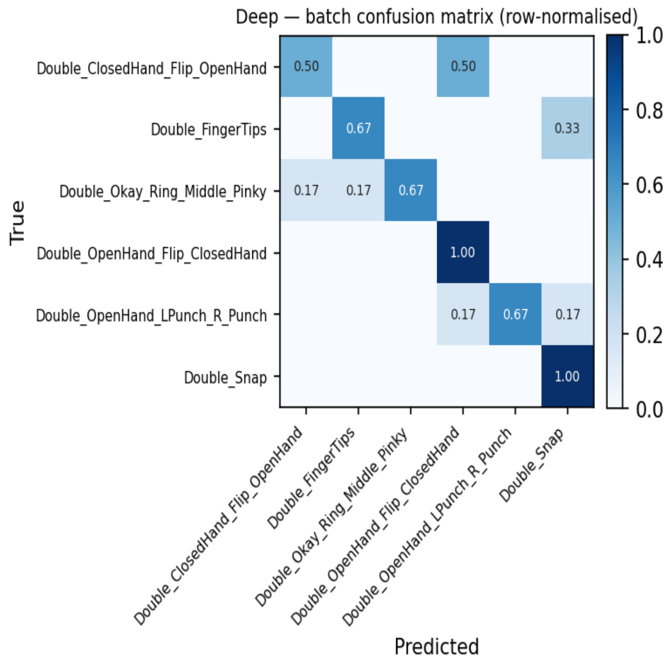
Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	3	6	0.500
Double_FingerTips	4	6	0.667
Double_Okay_Ring_Middle_Pinky	4	6	0.667
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	4	6	0.667
Double_Snap	6	6	1.000
Overall	**27**	**36**	**0.750**

Batch confusion matrix (8_Jestin)



Deep — batch confusion matrix (counts)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
3
3
4
2
1
1
4
6
1
4
1
6
0
1
2
3
4
5
6



Deep — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
0.50
0.50
0.67
0.33
0.17
0.17
0.67
1.00
0.17
0.67
0.17
1.00
0.0
0.2
0.4
0.6
0.8
1.0

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	53,094
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.6389 (23/36)

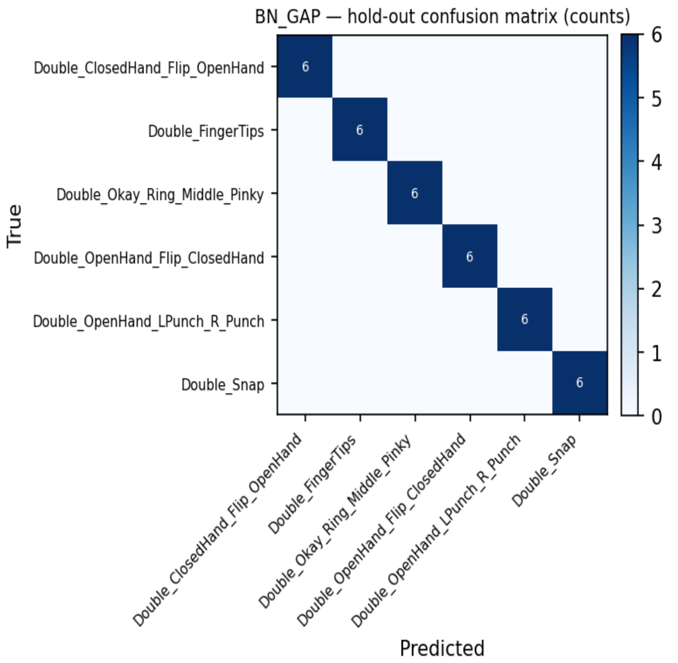
Cross-validation folds

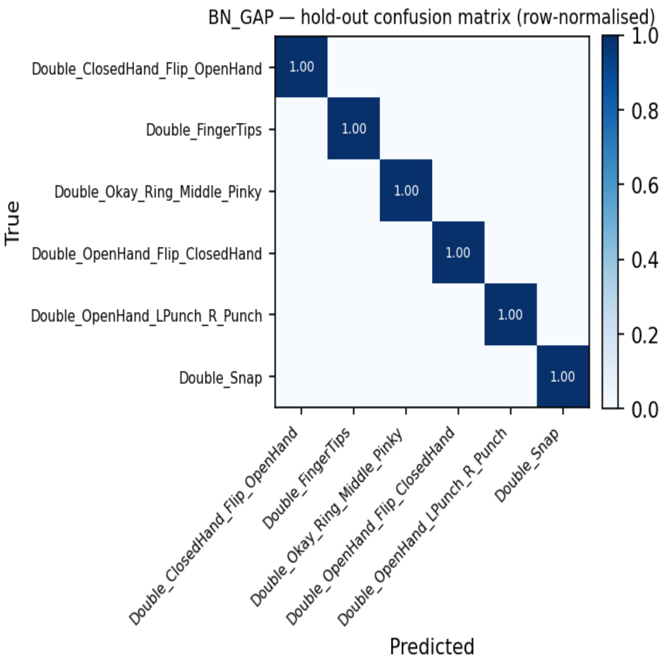
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

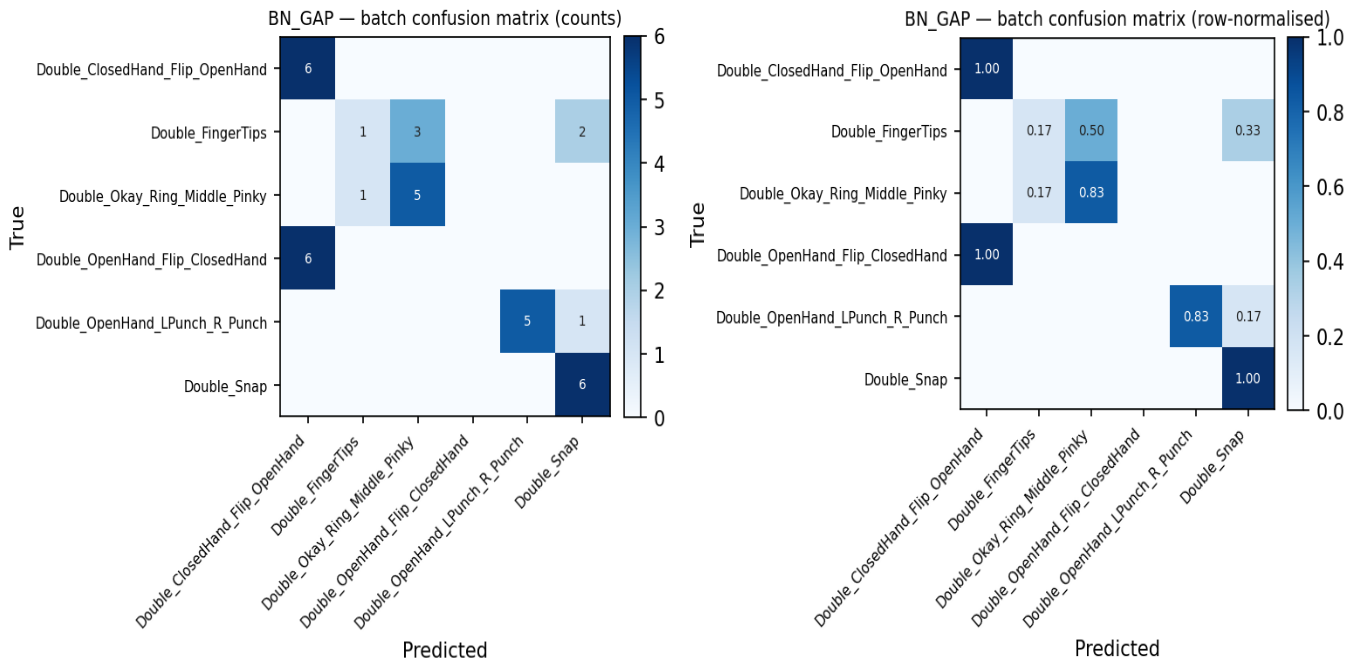




Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	6	6	1.000
Double_FingerTips	1	6	0.167
Double_Okay_Ring_Middle_Pinky	5	6	0.833
Double_OpenHand_Flip_ClosedHand	0	6	0.000
Double_OpenHand_LPunch_R_Punch	5	6	0.833
Double_Snap	6	6	1.000
Overall	**23**	**36**	**0.639**

Batch confusion matrix (8_Jestin)


BN_GAP — batch confusion matrix (counts)
BN_GAP — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	123,430
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.9167 (33/36)

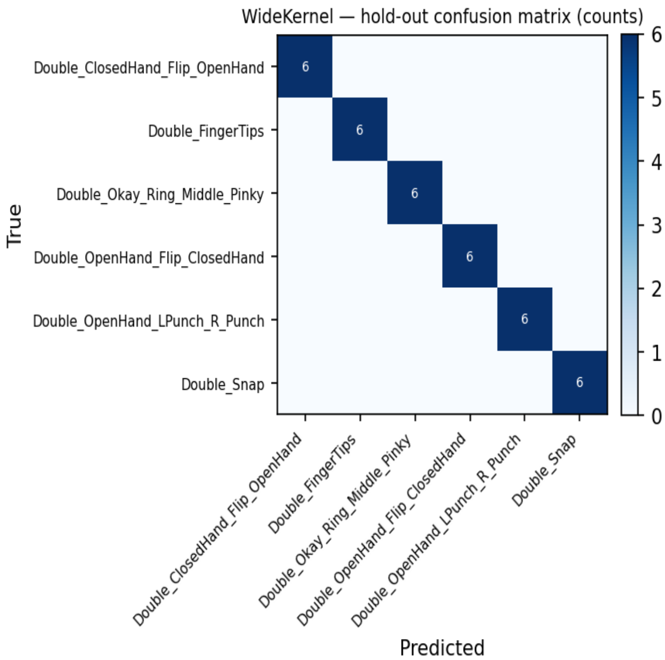
Cross-validation folds

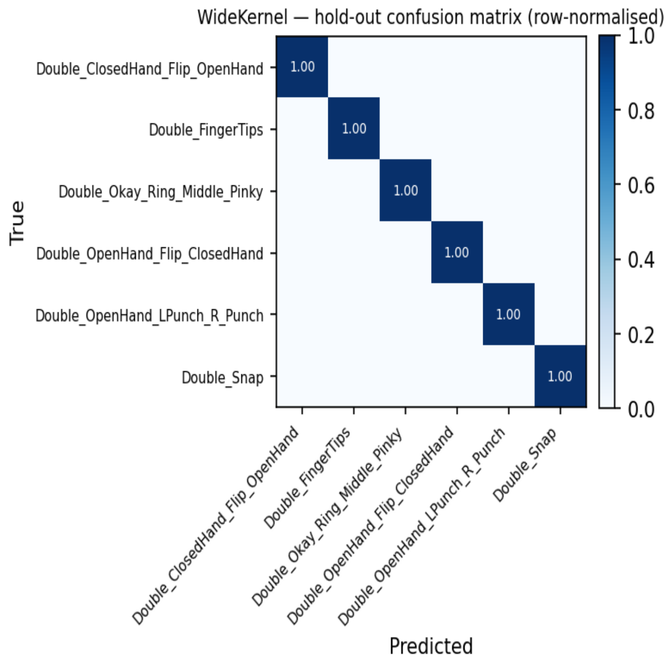
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	0.9783

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

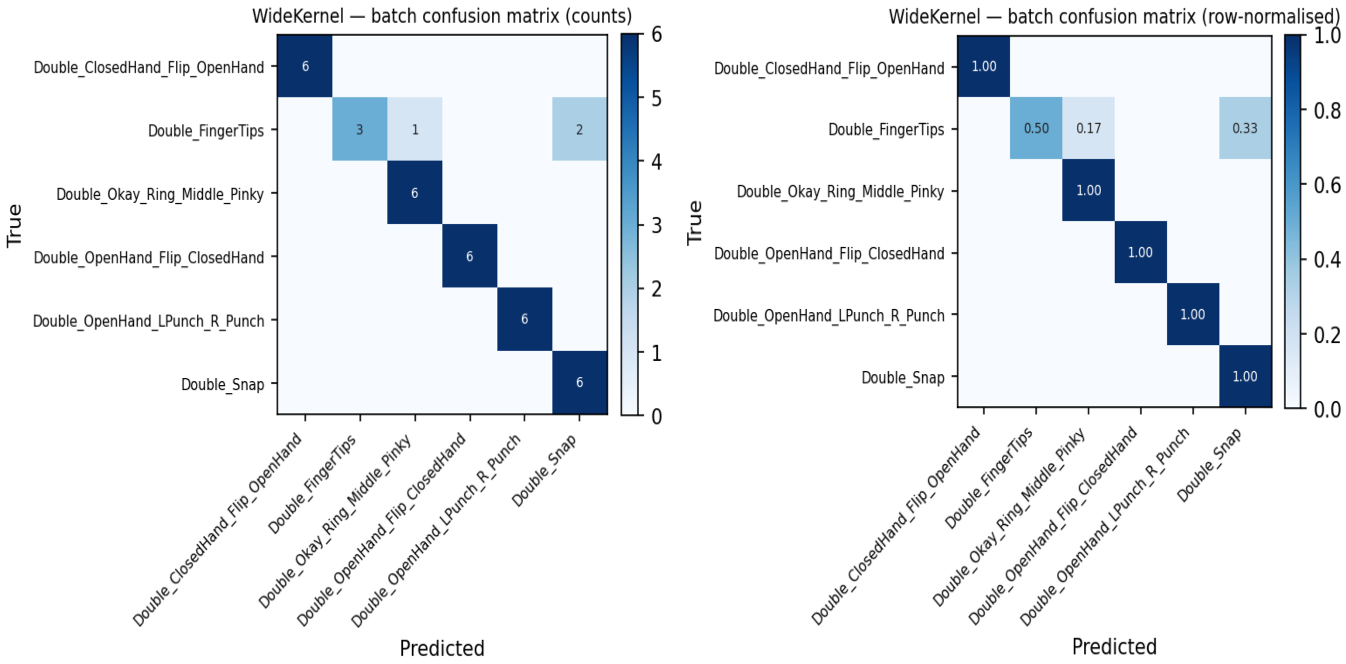




Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	6	6	1.000
Double_FingerTips	3	6	0.500
Double_Okay_Ring_Middle_Pinky	6	6	1.000
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	6	6	1.000
Double_Snap	6	6	1.000
Overall	**33**	**36**	**0.917**

Batch confusion matrix (8_Jestin)



WideKernel — batch confusion matrix (counts)
WideKernel — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	64,294
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.6667 (24/36)

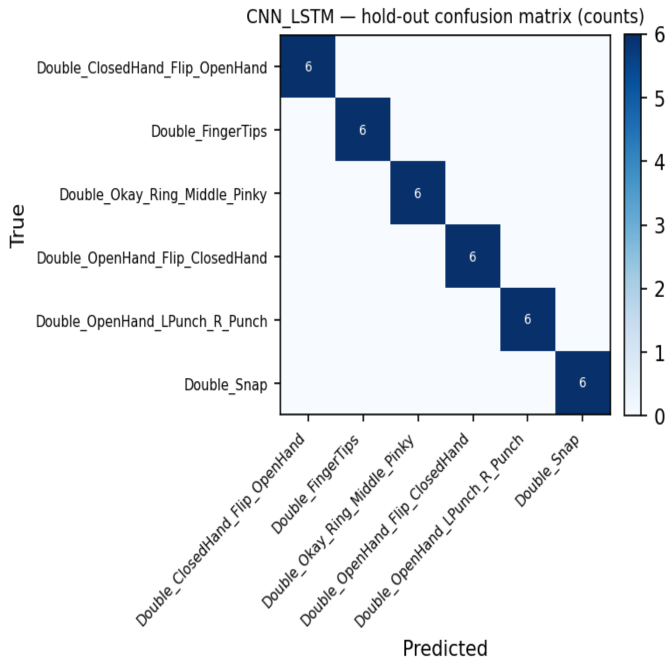
Cross-validation folds

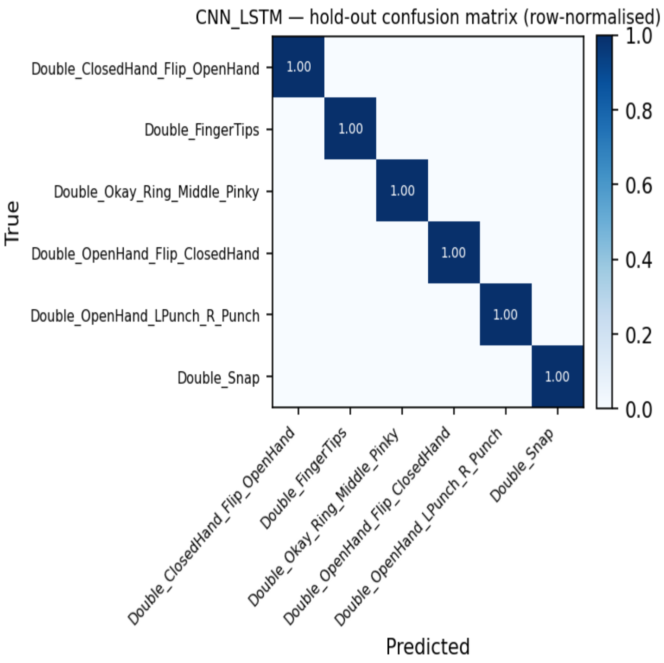
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

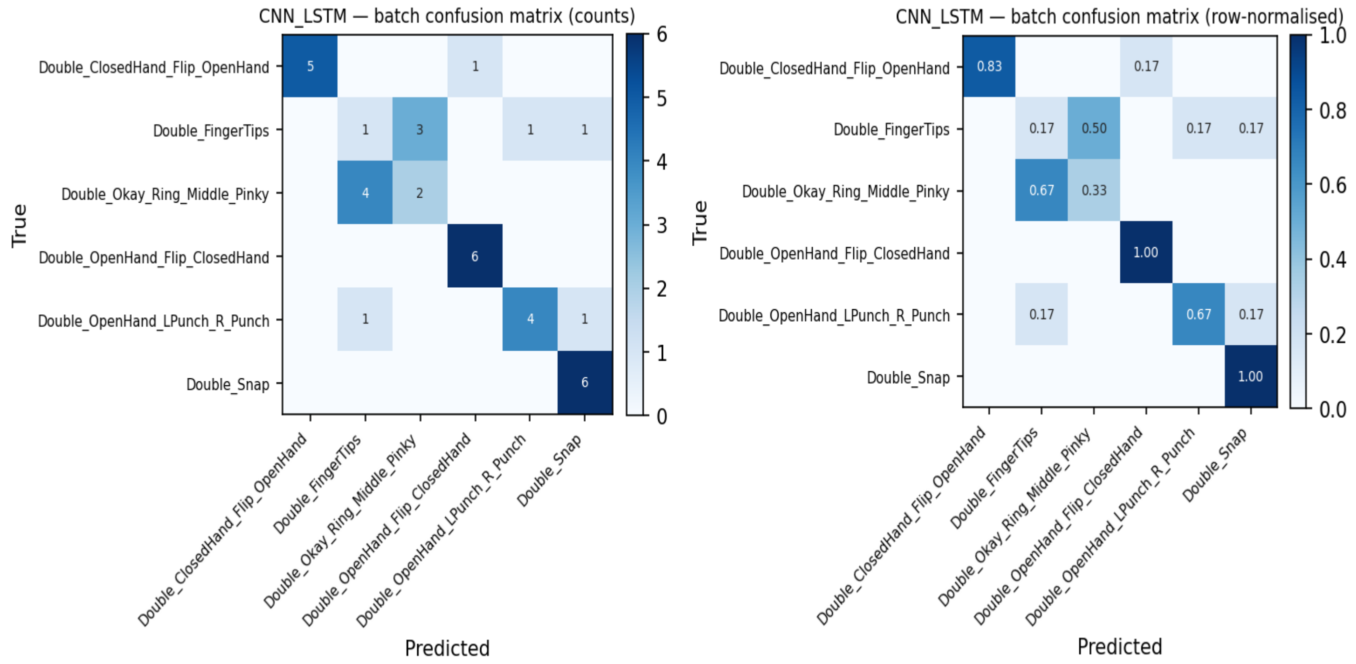




Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	1	6	0.167
Double_Okay_Ring_Middle_Pinky	2	6	0.333
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	4	6	0.667
Double_Snap	6	6	1.000
Overall	**24**	**36**	**0.667**

Batch confusion matrix (8_Jestin)


CNN_LSTM — batch confusion matrix (counts)
CNN_LSTM — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	101,414
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.6111 (22/36)

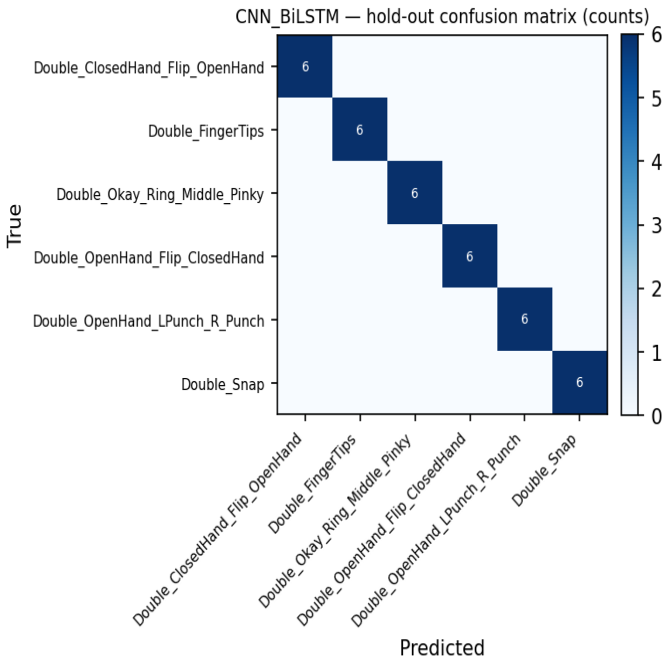
Cross-validation folds

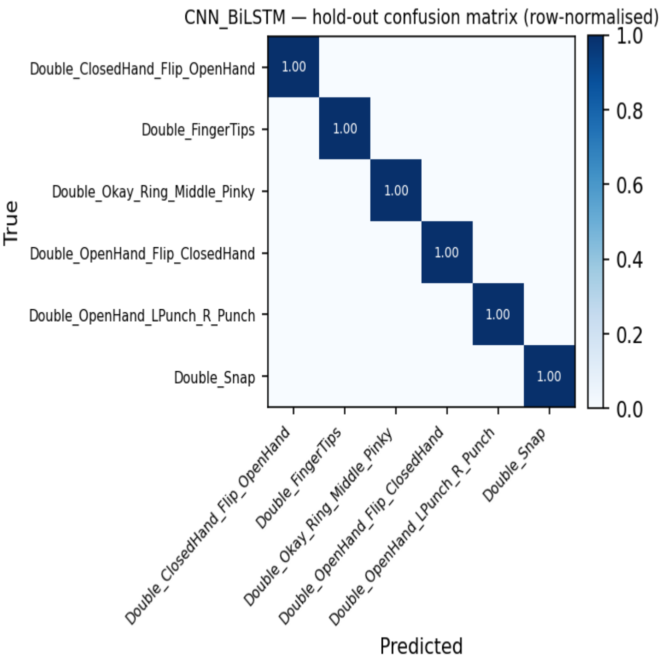
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

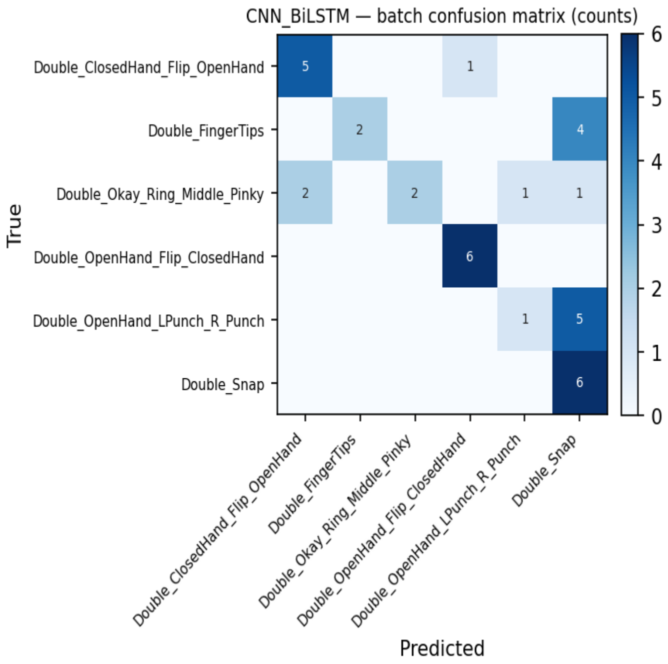


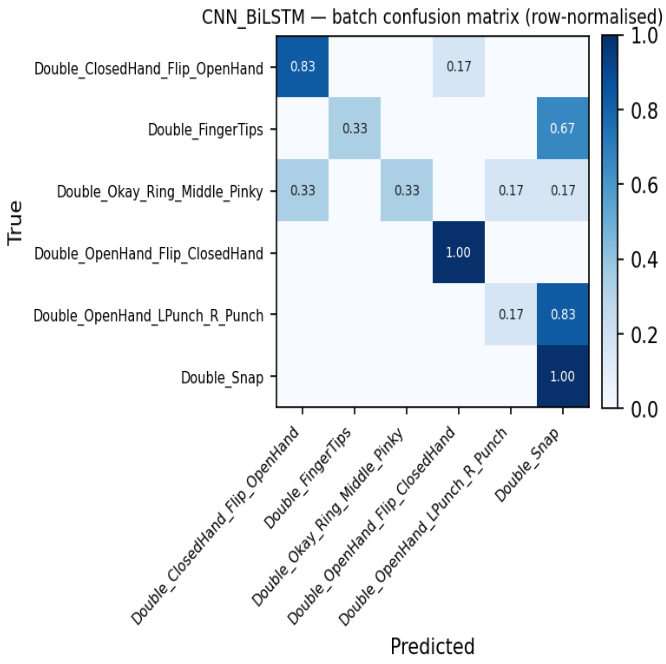


Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	2	6	0.333
Double_Okay_Ring_Middle_Pinky	2	6	0.333
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	1	6	0.167
Double_Snap	6	6	1.000
Overall	**22**	**36**	**0.611**

Batch confusion matrix (8_Jestin)


CNN_BiLSTM — batch confusion matrix (counts)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
5 1
2 4
2 2 1 1
6
1 5
6
0 1 2 3 4 5 6


CNN_BiLSTM — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
0.83 0.17
0.33 0.67
0.33 0.33 0.17 0.17
1.00
0.17 0.83
1.00
0.0 0.2 0.4 0.6 0.8 1.0

Variant: Model_I_1D

1-D adaptation of published Model I:
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

Trainable params	118,438
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.8056 (29/36)

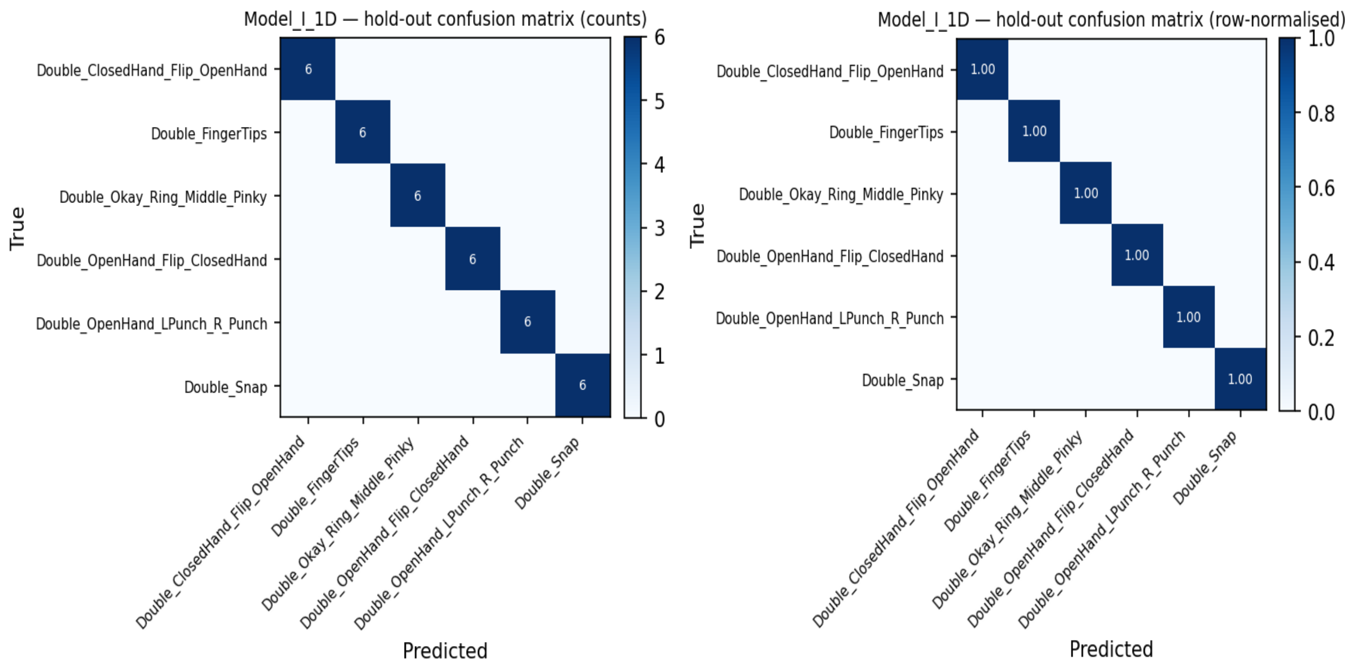
Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

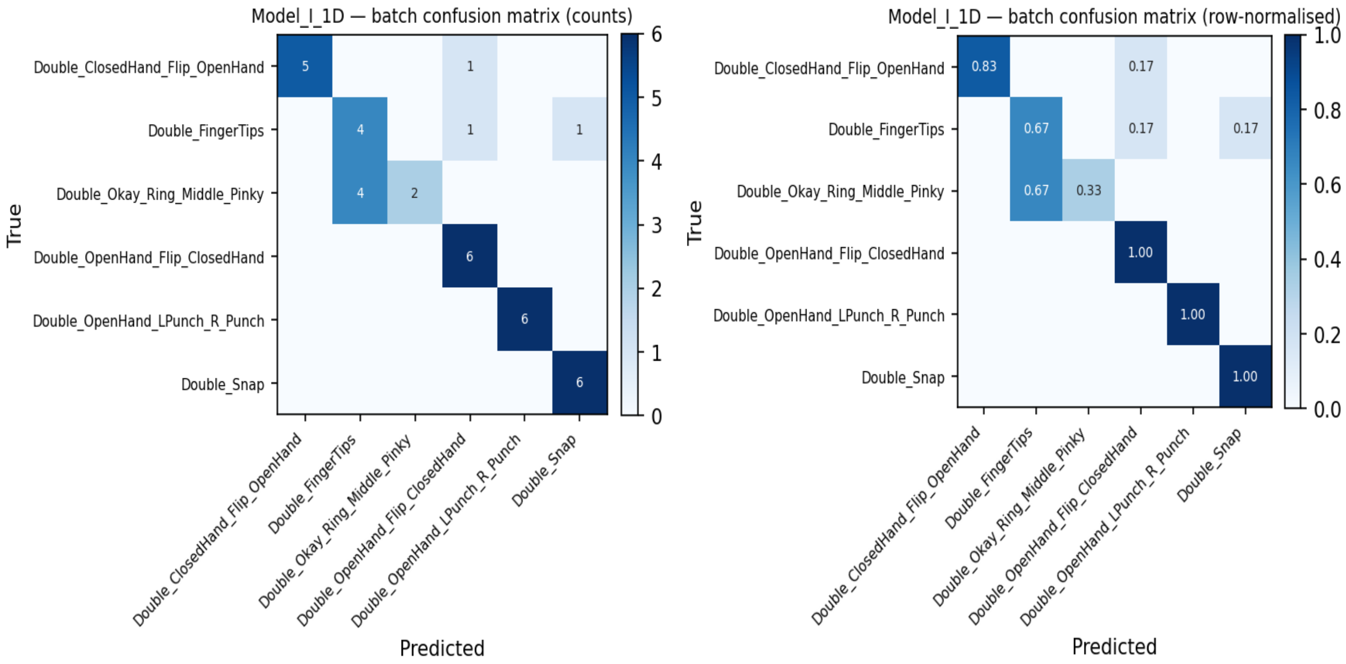
Hold-out confusion matrix



Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	5	6	0.833
Double_FingerTips	4	6	0.667
Double_Okay_Ring_Middle_Pinky	2	6	0.333
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	6	6	1.000
Double_Snap	6	6	1.000
Overall	**29**	**36**	**0.806**

Batch confusion matrix (8_Jestin)



Model_I_1D — batch confusion matrix (counts)
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
True
Predicted
Model_I_1D — batch confusion matrix (row-normalised)

Variant: Model_II_1D

1-D adaptation of published Model II (CNN+LSTM): Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

Trainable params	93,766
CV mean ± std	1.0000 ± 0.0000
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.6944 (25/36)

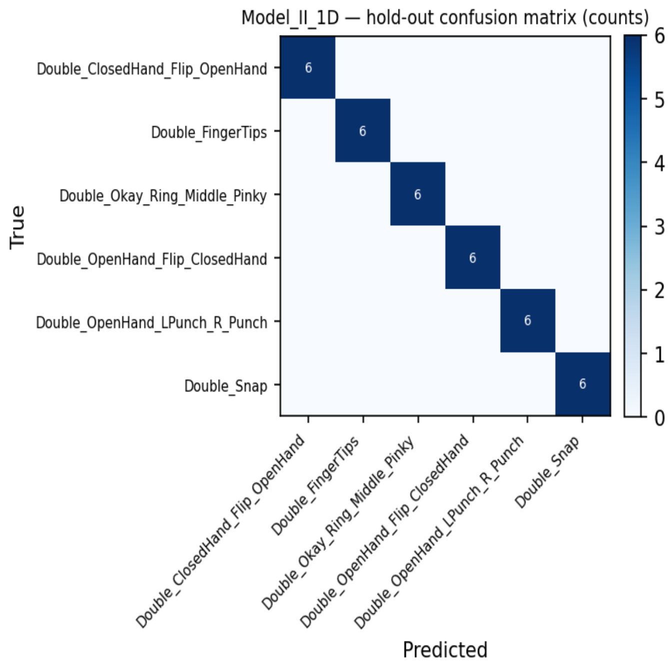
Cross-validation folds

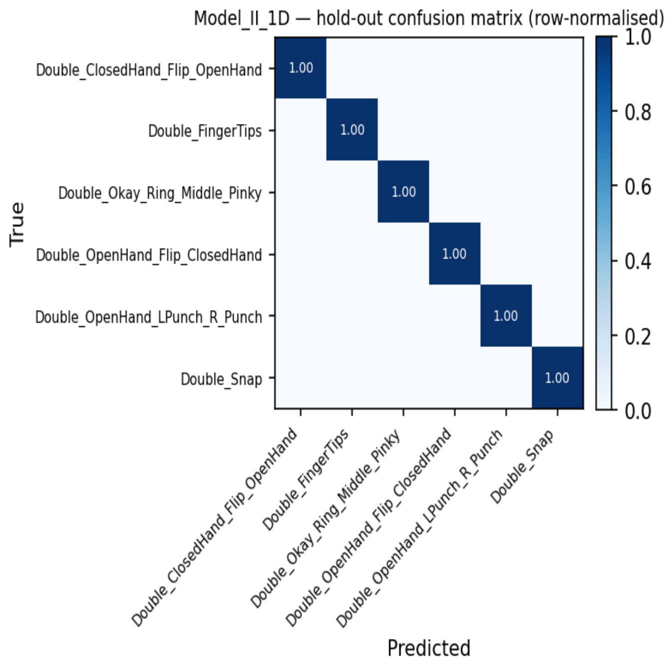
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1939	47	1.0000	1.0000	1.0000
2	1939	47	1.0000	1.0000	1.0000
3	1946	46	1.0000	1.0000	1.0000
4	1946	46	1.0000	1.0000	1.0000
5	1946	46	1.0000	1.0000	1.0000
6	1946	46	1.0000	1.0000	1.0000
7	1946	46	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	1.000	1.000	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Hold-out confusion matrix

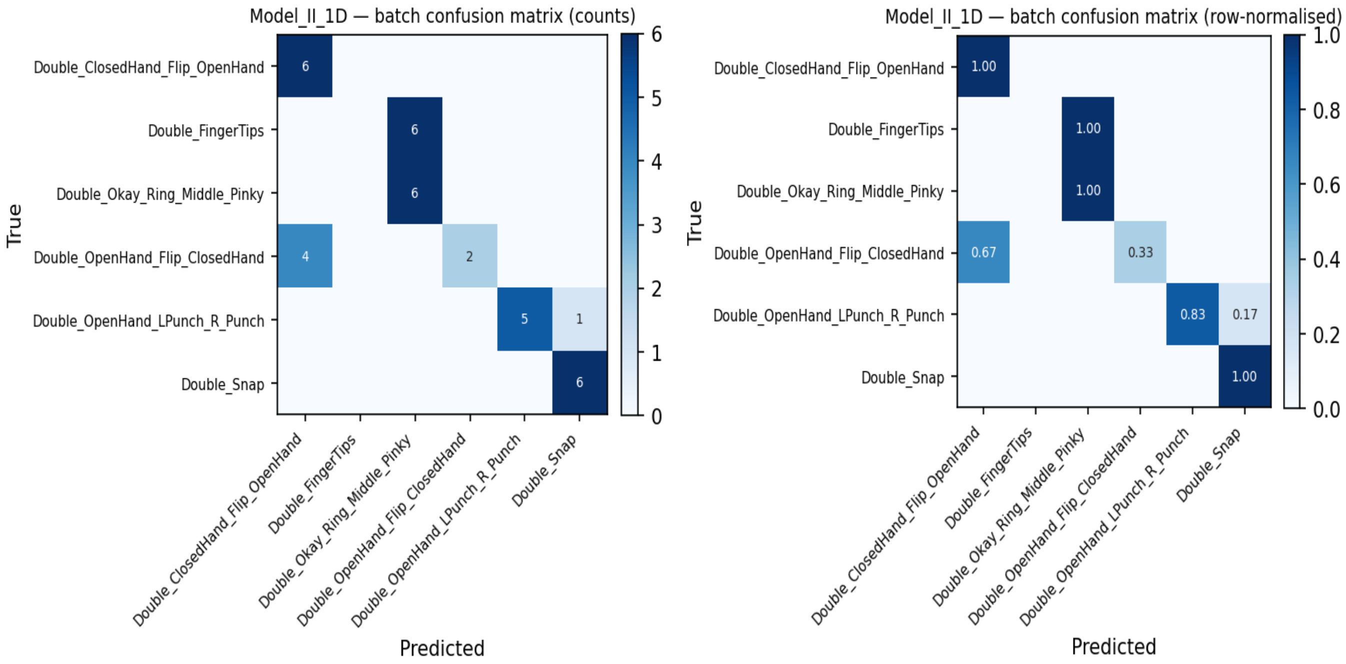




Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	6	6	1.000
Double_FingerTips	0	6	0.000
Double_Okay_Ring_Middle_Pinky	6	6	1.000
Double_OpenHand_Flip_ClosedHand	2	6	0.333
Double_OpenHand_LPunch_R_Punch	5	6	0.833
Double_Snap	6	6	1.000
Overall	**25**	**36**	**0.694**

Batch confusion matrix (8_Jestin)


Model_II_1D — batch confusion matrix (counts)
Model_II_1D — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap